Vishnu Krishnakumar Menon

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SUMMARY

AI/ML Engineer with 4+ years of experience developing production machine learning and generative AI systems, specializing in AI reliability, LLM evaluation, and scalable inference infrastructure. Experienced in architecting multi-agent workflows, hallucination detection systems, self-verifying RAG pipelines, and enterprise model-serving platforms. Supported systems handling 1M+ daily requests with 99.9% uptime and improved inference performance through retrieval optimization, model quantization, and caching. Focused on building reliable, evaluation-driven AI systems through LLM fine-tuning, retrieval optimization, automated verification, and production monitoring.

SKILLS

Programming & Libraries: Python (PySpark, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Pydantic), SQL, Hugging Face Transformers

Machine Learning & Deep Learning: XGBoost, Random Forest, Regression, Classification, Feature Engineering, Cross-Validation, A/B Testing, BERT, NLP, Model Fine-Tuning, QLoRA, DPO

Generative AI & Agentic AI: LLMs, LangChain, LangGraph, AI Orchestration, Prompt Engineering, Tool Calling, Model Context Protocol (MCP), Multi-Agent Workflows, Structured Outputs

RAG & Vector Search: Retrieval-Augmented Generation (RAG), OpenSearch, ChromaDB, Qdrant, Semantic Search, Embeddings, Retrieval Optimization, Reranking

LLM Evaluation & AI Reliability: Hallucination Detection, Faithfulness Evaluation, Claim Verification, Groundedness Evaluation, Prompt Regression Testing, Guardrails, Retrieval Quality Evaluation, Model Regression Testing

Data Engineering: Docling, Table Extraction, Apache Airflow, AWS Glue, ETL/ELT

MLOps & Deployment: Docker, Kubernetes, OpenShift, Terraform, MLflow, FastAPI, CI/CD, GitHub Actions, Model Registry, Model Monitoring, Automated Testing, Model Serving, Inference Optimization

Cloud & Databases: AWS (S3, EC2, SageMaker, Lambda, CloudWatch), Azure (Azure ML, AKS, Blob Storage), PostgreSQL, MySQL, MongoDB, Redshift

Observability & Tools: OpenTelemetry, Arize Phoenix, Prometheus, Grafana, Distributed Tracing, Git, Jira, Agile/Scrum

EXPERIENCE

C3 AI **Aug 2025 – Present |** Remote, USA

AI/ML Engineer

- Architected a multi-tenant LLM and RAG inference platform supporting 20+ enterprise deployments and 1M+ daily requests, separating retrieval, orchestration, and model-serving services on Kubernetes to maintain 99.9% uptime.
- Designed LangGraph-based AI orchestration workflows integrating document analysis, retrieval, LLM inference, and response verification, with automated validation gates, retry mechanisms, and fault isolation across enterprise deployments.
- Spearheaded the migration of 12 legacy ML workloads from on-premises infrastructure to AWS, containerizing services with Docker and managing cloud environments through Terraform; reduced infrastructure costs by 35% and deployment cycles from 10 days to 2 days.
- Orchestrated a cross-functional team of 6 engineers and 3 product owners to deliver a document-intelligence agent in 10 weeks, establishing evaluation criteria, milestone ownership, and phased production rollout procedures.
- Instituted automated model evaluation and drift-monitoring workflows using MLflow, Prometheus, and Grafana, integrating regression checks and monitoring alerts into CI/CD processes to detect model regressions 4x faster and reduce production incidents by 40%.
- Hardened embedding and vector-search infrastructure handling 50M+ documents through index sharding, response caching, and load-tested failover, maintaining p95 latency below 300 ms under 3x normal traffic.
- Mentored 4 junior engineers through weekly architecture reviews, production deployment walkthroughs, and code reviews, reducing their time to first production release from 8 weeks to 5.

Glean **Jan 2023 – Jul 2024 |** Bengaluru, India

AI/ML Engineer

- Optimized transformer-based semantic ranking models for enterprise search using hard-negative mining and embedding fine-tuning, improving offline top-3 relevance by 18% on NDCG and online click-through rates by 9%.
- Streamlined model development workflows through reusable Apache Airflow pipelines and standardized feature management, reducing training and evaluation turnaround from 5 days to 1.5 days.
- Investigated retrieval-quality regressions across 30+ enterprise data connectors, identifying index staleness and tokenization inconsistencies through query-level diagnostics and log analysis, restoring measured answer accuracy to 94%.
- Accelerated search inference performance through INT8 model quantization, request batching, and embedding caching, reducing p95 query latency by 40%, from 820 ms to 490 ms.

- Tuned RAG retrieval and generation configurations across a 1,500-question evaluation benchmark, experimenting with document chunk sizes, reranking, and citation-grounded prompting to reduce hallucinated responses by 27%.

Accenture | India

Data Scientist **Aug 2021 – Dec 2022 |** Kochi, India

- Constructed a retail demand-forecasting model using XGBoost and Prophet ensembles, engineering 60+ predictive features and improving forecast accuracy by reducing MAPE from 18% to 14%.
- Integrated an NLP-based service ticket classifier with the client's operational platform using FastAPI and Docker on Azure, automating 70% of ticket routing and saving approximately 1,200 agent hours quarterly.
- Automated weekly operational reporting for 5 business units using Python, SQL, and Airflow pipelines, replacing manual Excel processes and saving 25 analyst hours per week.
- Formulated a randomized experimentation framework for customer-retention initiatives, defining cohort assignments and statistical significance tests that supported a 6% improvement in retention across 40K customers.

Associate Data Scientist **Jan 2021 – Jul 2021 |** Kochi, India

- Prototyped a hybrid recommendation system combining collaborative and content-based filtering with Python, Pandas, and NumPy, achieving a 12% increase in click-through rates during A/B testing.
- Trained Random Forest and XGBoost churn prediction models, evaluating 40+ candidate features and performing cross-validated hyperparameter tuning to achieve an AUC of 0.86.
- Validated a BERT-based document-classification pilot on 10K labeled records, achieving a 91% F1 score through text preprocessing with SpaCy and NLTK and model version tracking in SageMaker.

EDUCATION

University at Buffalo, SUNY **Aug 2024 – Jan 2026**

M.S. in Computer Science and Engineering | GPA: 3.53/4.0

SCMS School of Engineering and Technology, Ernakulam, Kerala **Sep 2017 – Jun 2021**

B.Tech. in Electronics and Communication

CERTIFICATION

AWS SimuLearn - AI Practitioner - Training Badge
AWS Certified Machine Learning Engineer Associate - (MLA-C01)
AWS Certified AI Practitioner - (AIF-C01)
DeepLearning.AI: Generative AI with LLMs, RAG
Anthropic: Claude Code 101, Introduction to MCP

PROJECTS

TheraMind: Dual-Mode Clinical LLM Agent | *Llama 3.2-3B, QLoRA, DPO, ChromaDB, FastAPI, Docker*
vkm47-theramind-ai.hf.space/ | github.com/Vishnu-K-Menon/TheraMind

- Built and deployed a clinical LLM agent fine-tuned from Llama 3.2-3B using QLoRA and DPO on 9,250 clinical dialogues, improving ROUGE-L from 0.40 to 0.78 with a 19.5-minute training runtime on an RTX 4090.
- Developed a ChromaDB-based RAG architecture supporting clinical intake and note-generation workflows, combining retrieval-grounded responses, safety prompts, constrained decoding, and prompt-injection testing to reduce cross-mode contamination risks.
- Applied Direct Preference Optimization with $\beta = 0.1$ using 49 preference pairs derived from model errors and reference responses, refining response alignment and separating preference-based tuning from standard supervised fine-tuning.

LEDGER: Self-Verifying RAG | *Docling, OpenTelemetry, Arize Phoenix, AWS EC2*
github.com/Vishnu-K-Menon/LEDGER

- Developing a self-verifying RAG system for U.S. federal budget and energy reports that decomposes generated answers into atomic factual claims, checks each claim against retrieved source evidence using a local 7–8B model, and rewrites or removes unsupported statements.
- Diagnosed a Docling table-parsing defect affecting 5.5% of table cells and implemented PDF text-layer reconstruction, raising held-out table-cell accuracy from 30.5% to 100% on evaluated Economic Report of the President tables.
- Implemented distributed tracing with OpenTelemetry and a self-hosted Arize Phoenix instance on AWS EC2, alongside a crash-safe, resumable Batch API evaluation runner and 223 automated tests integrated into GitHub Actions CI.